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Convergent Evolution of Pregnancy in Vertebrates

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Keywords

brooding, gestation, live birth, oviparity, placenta, viviparity

Abstract

Viviparity (live birth) represents a significant evolutionary innovation that has emerged in hundreds of lineages of invertebrate and vertebrate animals. The evolution of this trait from the ancestral state of egg laying has involved complex morphological, behavioral, physiological, and genetic changes, which enable internal development of embryos within the female reproductive tract. Comparable changes have also occurred in oviparous, brooding species that carry developing embryos in locations other than the female reproductive tract. This review explores the taxonomic distribution of vertebrate viviparity and brooding (collectively termed pregnancy), discusses the adaptations associated with internal incubation, and examines hypotheses surrounding the evolution of pregnancy in different lineages. Understanding the mechanisms that have led to the emergence of this trait can illuminate questions about the evolution of reproductive complexity and the processes that led to the emergence of evolutionary innovations that have shaped the remarkable diversity of Earth's fauna.

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INTRODUCTION

One of the most striking examples of convergent evolution in animals is viviparity (live birth). This trait has evolved from the ancestral state of oviparity (egg laying) at least 150 times in vertebrates (1) and many times in invertebrates (2). In contrast, other well-known examples of convergence in vertebrate evolutionary innovations have evolved far fewer times—for example, powered flight has evolved only three times (1). The recurrent origins of viviparity across diverse taxa present an excellent opportunity to understand more about evolutionary processes related to viviparity and innovations in general.

For example, it is possible to conduct robust tests of hypotheses surrounding the conditions under which live birth is favored and the repeatability of the mechanisms that support internal embryo development. Such studies can shed light on the mechanisms facilitating evolutionary innovation, providing the phylogenetic replication that is impossible with other phenotypes, such as lactation, which has evolved only once (e.g., 3). Adding to the utility of viviparity in studying broad evolutionary mechanisms is that some extant species exhibit a transitional reproductive form, intermediate between oviparity and viviparity. Observing transitional forms of complex traits is rare outside the fossil record. Transitional forms of viviparity allow us to watch evolution in action and assess how quantitative changes to physiology, morphology, behavior, genetics, and epigenetics contribute to the evolution of novel phenotypes (4).

VIVIPARITY, BROODING, TRANSITIONAL FORMS OF PREGNANCY, AND PLACENTAE

Terminology

Oviparity is the reproductive state (also known as parity mode) referring to the deposition of either (*a*) unfertilized eggs (which are later fertilized externally) or (*b*) eggs that are fertilized internally and then laid within an egg covering (5). In oviparity, a substantial amount of embryonic development, or all embryonic development in some species, occurs in the external environment. Viviparity describes the incubation of embryos within the female reproductive tract until they have completed development (5). Although historically the term ovoviviparity has been used, it has many different contradicting definitions, and its use is discouraged (e.g., 2, 5–7).

In viviparity, eggs are fertilized inside the female reproductive tract, and the site of gestation is either the oviduct (uterus), ovarian lumen, or ovarian follicle, depending on the species (1). Oviparity with brooding is a strikingly similar reproductive strategy to viviparity, in which embryos develop in or on the parent's body at sites other than the female reproductive tract (e.g., 2, 8, 9). The site of gestation varies across oviparous brooding species. For example, some anurans brood embryos in skin pockets or chambers (e.g., families Amphignathodontidae, Cryptobatrachidae, and Hemiphractidae) (reviewed in 9); aspredinid catfish brood embryos attached to cotylephores (vascularized skin outgrowths) (10); some brooding bivalves incubate embryos in modified gill structures (2); and male syngnathid fishes incubate embryos in specialized ventral skin patches or pouches (reviewed in 8). The term pregnancy can encompass both viviparity and oviparity with brooding.

Transitional Forms of Pregnancy

Intermediate phenotypes between oviparity and viviparity in vertebrates are very rare but have been well established in three lizards: *Lerista bougainvillii* (Bougainville's skink, found in Australia), *Saiphos equalis* (three-toed skink, found in Australia), and to some extent *Zootoca vivipara* (common lizard, found across much of Eurasia) (4). In these species, three different reproductive phenotypes

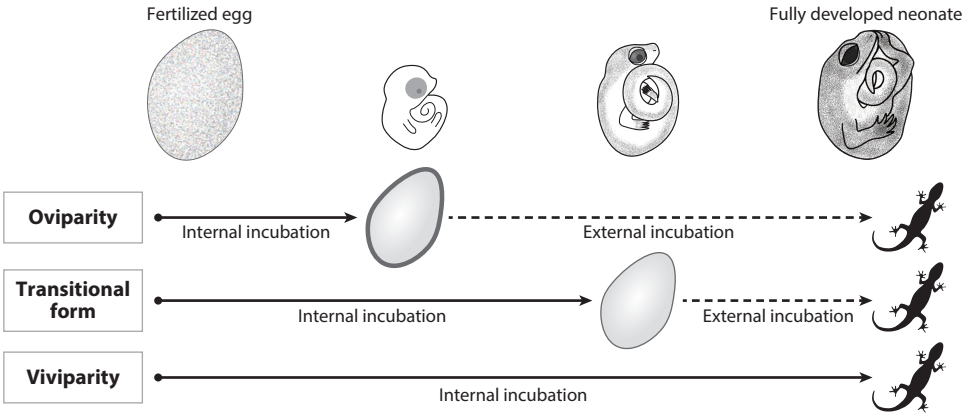

Figure 1

Transitional forms of pregnancy have been observed in three extant lizard species. This phenotype appears intermediate between oviparity and viviparity, with thinly shelled eggs deposited late in embryonic development and incubated externally for a short period before hatching. Figure adapted with permission from Whittington et al. (4) (CC BY-NC-ND-4.0).

(oviparity, viviparity, and a transitional form of pregnancy) are observed repeatedly at different geographic locations, and the phenotype appears to be genetically encoded. Viviparity has evolved recently in these taxa, e.g., ~4.5 Mya in *Z. vivipara* (11) and 0.01 Mya in *L. bougainvillii* (12). All three lizards have become important models for understanding the evolution of viviparity because of their recent origins of this trait and because they allow comparisons between parity modes that are not confounded by speciation (e.g., 13–16).

In transitional forms of squamate pregnancy, compared to oviparity, embryos are retained for an extended period inside the mother's body and achieve an advanced stage of development before laying. They are laid inside thinly shelled or partially calcified eggs that hatch after a short period of external incubation (reviewed in 4) (**Figure 1**). Transitional forms are considered rare because they incur the costs of both oviparity and viviparity but receive only some of the advantages (e.g., 13, 17–20). For example, transitional females are physically burdened for extended periods as in viviparity and require a suitable nest site as in oviparity. Transitional females can thermoregulate embryos to optimize development, but not for the entirety of development as in viviparity. Semi-calcified eggs, often laid by transitional females, may be more susceptible to fungal infection, desiccation, and predation than oviparous eggs with thicker shells. Such transitional forms may, therefore, represent a fitness valley between oviparity and viviparity.

There are also several vertebrates with less well-established models of transitional forms of pregnancy. Some rockfishes (*Helicolenus* spp.) vary in the developmental stages at which they deposit their embryos (21). Such variation in how long embryos are retained inside the female reproductive tract is also present in some catsharks (*Halaelurus* spp.) (e.g., 22, 23). The phenotypic variation in reproductive strategy present in these genera makes them valuable models for studies of viviparity evolution, but they have been generally overlooked compared to the transitional forms of squamate pregnancy.

Placentation

Placentae are present in many viviparous and oviparous, brooding species. A placenta is defined as any "intimate apposition or fusion of the fetal organs to the maternal (or paternal) tissues for

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physiological exchange" (24, p. 156). The placenta is a transient organ, present only during pregnancy, and is critical in transporting molecules between parent and embryo. Depending on the species, the molecules transported include respiratory gases, nutrients, water, immune factors, and wastes; placentae may have additional functions (e.g., endocrine). Like viviparity and brooding, placentae also have evolved repeatedly in animals; for example, the complex nutritive placenta (which transports significant quantities of nutrients from parent to embryo) has evolved at least 16 times in vertebrates (25) (see the section titled Transport of Nutrients from Parent to Embryo).

The morphology and physiology of placentae vary across species; for discussion of these topics, readers are directed to recent in-depth reviews (e.g., 1, 25–27). Briefly, placentae can be composed of parental tissues including the uterus, ovary, or skin; embryonic contributions to placentation include gill and pharyngeal epithelium, hindgut extensions, chorioallantois, yolk sac, skin, or pericardium (reviewed in 25) (**Figure 2**). The intimacy of the connection between parent and embryo at the placental interface also varies and can be described based on the number of interhemal barriers between embryonic and parental vascular systems. This classification system was developed originally for mammalian placentae (28) and includes (*a*) the two tissue types being simply apposed to one another (epitheliochorial), (*b*) erosion of parental epithelium such that the embryonic tissues are apposed to stromal tissues (syndesmochorial), (*c*) erosion of parental tissues such that the embryonic tissues are apposed to parental blood vessels (endotheliochorial), and (*d*) parental tissues eroded completely such that the embryonic tissues are bathed in parental blood (hemochorial). Various eutherian groups, such as great apes (including humans), rats, and mice, display the latter, most intimate placental connection. Regardless of the invasiveness of placentation or the tissue types forming placentae, molecular transport across these organs is very efficient due to the thinning of tissues between the two circulatory systems and increased vascularity and surface area of contact at the placental interface (25).

CONVERGENT EVOLUTION OF VIVIPARITY ACROSS ANIMALS

Every major lineage of vertebrates contains viviparous taxa, except for testudines, crocodilians, and birds. Viviparity also appears in at least 160 invertebrate families (2; this analysis defined viviparity as embryonic development inside the parent at sites including ovary, sexual duct, and body cavity), including some ascidians and salps, as well as some platyhelminths, arthropods, bryozoans, sponges, nematodes, echinoderms, molluscs, annelids, ribbon worms, cnidarians, dicyemids, spiny-headed worms, velvet worms, orthonectids, acoelomorphs, cycliophorans, rotifers, gastrotrichs, and loriciferans. The number of origins of viviparity in invertebrates has yet to be determined. There may be viviparous species that are yet to be identified, and further research is required to clarify the evolutionary history of many lineages, including identifying whether viviparity has evolved multiple times within some groups.

In contrast, we have reasonable estimates of the number of origins of vertebrate viviparity [>150 (1)], the focus of this article. These estimates are founded on the strongly supported hypothesis that the basal amniote was oviparous. Although a recent analysis highlighted the potential that the ancestral amniote was viviparous (29), we suggest that more biological evidence is required to definitively support this argument against the longstanding paradigm of basally oviparous amniotes. Analysis of the vertebrate phylogeny shows striking patterns of transitions from oviparity to viviparity, including an exceptional number of independent origins of viviparity in squamate reptiles (snakes, lizards, and amphisbaenians) (**Figure 3**).

Almost every extant mammal is viviparous, incubating embryos inside the uterus (modified oviduct, which originates developmentally from the Müllerian duct). Although mammalian viviparity is of considerable interest in fields including agriculture and medicine, conducting

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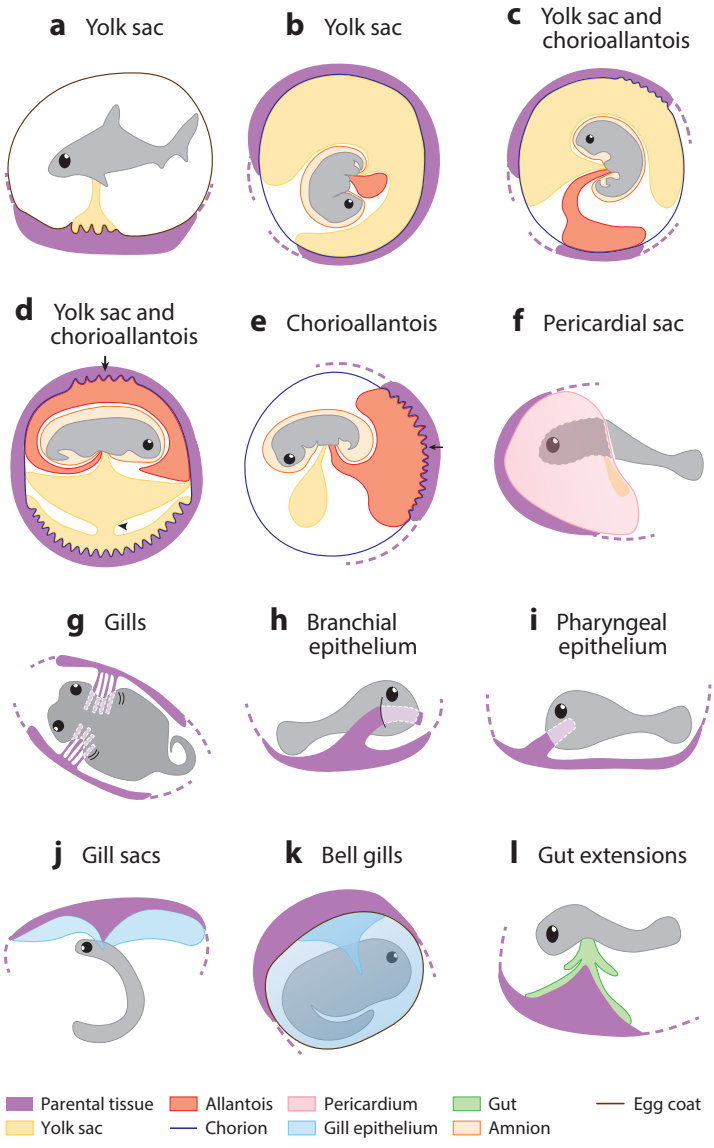

Figure 2

Placentae have evolved in a range of viviparous and oviparous, brooding taxa. The placenta is composed of parental tissue (e.g., uterus, ovarian tissue, skin; indicated here in *purple*) closely apposed to various embryonic tissues, functioning to exchange molecules between parent and embryo. The embryonic portion of the placenta can be derived from a range of different tissues, depending on the species. Examples are yolk sac (*yellow*), allantois (*red*), chorion (*dark blue*), pericardium (*pink*), gill epithelium (*blue*), and gut (*green*). Brown indicates egg coat; orange denotes amnion. The panels indicate example placentae in (*a*) *Rhizoprionodon taylori* (shark), (*b*) *Setonix brachyurus* (marsupial), (*c*) *Phascolarctos cinereus* (marsupial), (*d*) *Pseudemoia entrecasteauxii* (arrow indicates placentome, arrowhead indicates yolk cleft), (*e*) generic eutherian with chorioallantoic placenta, (*f*) *Heterandria formosa* (teleost), (*g*) *Gymnura micrura* (ray), (*h*) *Jenynsia lineata* (teleost), (*i*) *Zoarces viviparus* (teleost), (*j*) *Typhlonectes compressicauda* (amphibian), (*k*) *Gastrotheca riobambae* (amphibian), and (*l*) *Ameca splendens* (teleost). Figure adapted with permission from Reference 25.

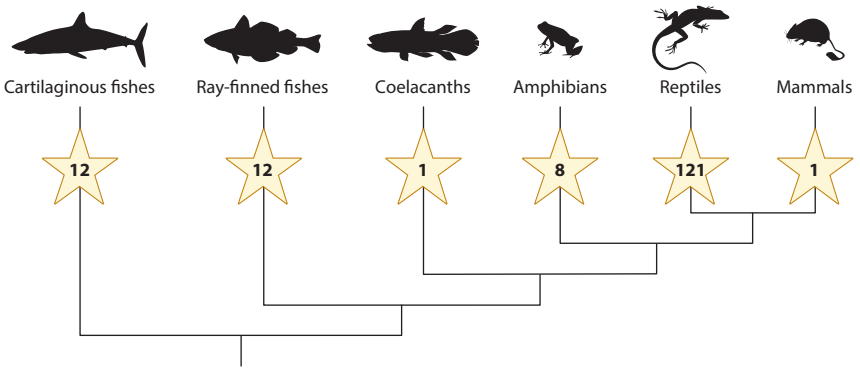

Figure 3

Viviparity has evolved independently more than 150 times in vertebrates. The phylogeny illustrates major vertebrate clades that contain viviparous taxa, indicating the number of origins of viviparity for each lineage, using data from Blackburn (1) and Blackburn & Hughes (35); an additional origin of viviparity in extinct placoderms is not shown. Adapted with permission from Whittington (151).

studies of the evolution of viviparity in this group is challenging because of the ancient origin of viviparity in this lineage. Oviparous monotremes and viviparous therians diverged 191–163 Mya (30), precluding comparative studies of closely related oviparous and viviparous taxa to elucidate the mechanisms underpinning parity mode differences. Ideally, we would study taxa in which viviparity is of recent origin: Squamate reptiles are ideal. In squamates, viviparity has evolved frequently (at least 115 origins) (1) and recently (e.g., as recently as 10,000 years ago; 12). Twenty-seven families of snakes and lizards contain oviparous members (31, 32). The incubation site is the uterus, and squamate placentae function to transport respiratory gases, water, and calcium (33); there are also six origins of substantial placental nutrient transport, all in scincid lizards (1). Squamates are also the only vertebrates to exhibit within-species variation in reproductive mode. There is reliable evidence that ten species exhibit this phenomenon (34); populations of three of these species also display transitional forms of pregnancy (4).

Viviparity has also evolved frequently in chondrichthyans (sharks, skates, rays, sawfish, chimaeras); 55% of extant members of this lineage are viviparous, and this trait has evolved independently at least 12 times (35), including in every order of sharks except Heterodontiformes (reviewed in 36). In chondrichthyans, embryos are gestated within the uterus. The mechanisms to supply embryonic nutrition are exceptionally diverse in this group (see the section titled Transport of Nutrients from Parent to Embryo below).

Approximately 2.5% of extant teleost modern bony fishes (Actinopterygii: Teleostei) are viviparous (1, 22), representing at least 11 independent origins of viviparity [there is an additional origin of viviparity in extinct non-teleostean actinopterygian *Saurichthys* (37)]. In contrast to amniotes and chondrichthyans, teleosts lack Müllerian ducts and, therefore, have no oviducts in which to incubate embryos (38). Instead, the site of gestation is the lumen of the ovary or inside the ovarian follicle. Like chondrichthyans, viviparous teleosts have diverse methods to supply nutrition to embryos during pregnancy. Viviparity is also present in coelacanths (Sarcopterygii: Coelacanthiformes). In these fish, embryos are incubated in the right uterus (the left is nonfunctional; 39) and possibly nourished by a placenta or oophagy (egg eating; see the section titled Transport of Nutrients from Parent to Embryo), although these facts remain to be confirmed (39, 40).

Finally, among extant vertebrates, Lissamphibia (extant amphibians) represent eight origins of viviparity; 1% are viviparous (1). Viviparity has evolved twice in frogs and toads, twice in urodeles,

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and four times in caecilians. The site of gestation is the uterus, and the products of gestation vary from tadpoles (larviparity) to metamorphosed neonates (pueriparity) (9). As in chondrichthyans and teleosts, viviparous lissamphibians provide embryonic nutrition during gestation via various mechanisms (1) (see the section titled Transport of Nutrients from Parent to Embryo).

The focus above is on the origins of viviparity in extant vertebrates. However, the fossil record preserves evidence of many examples of viviparity in extinct species. The oldest record of viviparity is a 380-million-year-old fossil of the placoderm *Materpiscis attenborough*i, which even preserves evidence that this species possessed a placenta (41). Other examples of extinct viviparous taxa include the holocephalan *Harpagofututor volsellorhinus* (42); the actinopterygian *Saurichthys* (37); various mesosaurs, pachypleurosaurs, notosaurs, plesiosaurs, ichthyosaurs, choristoderans, and mosasaurs (all aquatic reptiles); and the lizard *Yabeinosaurus tenuis* (42 and references therein). It is noteworthy that five of the fossil origins of viviparity are in aquatic reptiles, because we know that extant amniote eggs cannot develop underwater, and all extant amniotes either return to land to lay eggs or are viviparous (43). Fossils of other aquatic amniotes may yield discoveries of further viviparous taxa.

WHY HAVE VIVIPARITY AND INTERNAL BROODING EVOLVED?

The diversity of evolutionary histories and habitats of viviparous taxa make a single explanation for the evolution of viviparity across all species unlikely. However, hypotheses have been developed for specific taxonomic groups [e.g., invertebrates (44), reptiles (45–47), and fish (22)]. The extent to which empirical evidence supports these hypotheses is mixed. Most efforts have focused on lineages with multiple origins of viviparity (e.g., squamates). The absence of a suitable level of phylogenetic replication, as is the case with mammals, of which the only oviparous lineage is monotremes, makes meaningful investigation challenging in many groups.

Determining the costs and adaptive benefits of viviparity is fundamental to understanding why viviparity evolves. Relative to oviparity, viviparity appears to impose more restrictions on female reproductive output and incurs a greater cost per reproductive event (48–50). Yet viviparity must increase relative fitness in some situations because it has evolved multiple times and persisted over millions of years of evolutionary history (46). Possible fitness benefits of viviparity include reducing embryonic exposure to predation or mitigating the effects of environmental instability on embryonic development (45, 46, 51). Viviparity has likely enabled the exploitation of new ecological niches for squamates (e.g., cold climates; 52), amphibians (terrestrial environments; 53), and fishes (pelagic environments; 22).

Squamates are the primary model systems for hypothesis generation for the evolution of viviparity. The cold climate hypothesis (CCH) is the leading hypothesis in squamates and argues that extended retention of developing embryos inside the mother evolves in cooler environments, as mothers can better shield internally gestated embryos from cold extremes and provide faster rates of embryogenesis relative to eggs oviposited in a nest (45). This increased development rate allows embryos to emerge earlier and capitalize on a longer growth period before winter, likely increasing the chances of offspring surviving their first winter (45). Recently, the role of oxygen limitation in high-elevation environments has been raised as an additional factor in driving the transition to viviparity in the cold (hypoxia hypothesis; reviewed in 54).

Modern large-scale data sets have allowed powerful comparative studies validating observed associations (46) between cold climates and increased prevalence of viviparity (52, 55). Moreover, ample experimental manipulations support viviparity outperforming oviparity in cold regions (56, 57). Mechanistic models are a powerful tool in viviparity research because they link the proximate and ultimate causes of viviparity and its benefits in cold climates, validating the assumptions

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that viviparity results in faster development and lower energetic costs in these environments (58). The ready availability of large trait-based data sets and more user-friendly methods of mechanistic modeling (e.g., NicheMapR; 59) provide an exciting tool set for biologists to test competing hypotheses about viviparity and climate at a global scale.

Whereas the CCH focuses on suboptimal external thermal conditions during development, several other hypotheses within squamates and other taxa focus on competing aspects of nest quality. The nest availability or dry climate hypothesis considers the absence of suitable nesting sites as a driver of the evolution of viviparity (60, 61). Although conceptually sound in squamates (46) and invertebrates (44), there is negligible empirical support for it. For amphibians, the nest availability hypothesis is considered the leading hypothesis, with empirical support from correlative analyses of parity mode and proxies of nest quality (53, 60). The related environmental predictability hypothesis argues that mothers retain embryos under conditions with variable nest quality, such as environments prone to flooding, thermal extremes, unpredictable thermal conditions, or desiccation (45, 46). Empirical testing of this hypothesis is limited and unsupportive so far (52, 62–64). Although not mutually exclusive of other hypotheses, decreased embryonic predation compared to oviparity is often cited as an adaptive benefit of viviparity (22, 46, 65). This theory has little empirical support, and further investigation is required.

The maternal manipulation hypothesis (MMH) extends the CCH in squamates and presents viviparity as an adaptation that allows mothers to exert more control over offspring fitness during development (66). Underlying the MMH are the assumptions that mothers will shift thermoregulatory preferences and increase thermoregulatory precision while pregnant (66). Indeed, there is strong evidence of maternal thermoregulation influencing locomotor (67, 68) and other fitness-relevant traits in offspring (69, 70). Experimental studies generally support the assumption that mothers alter thermal preferences during pregnancy, though this is not always the case (68, 70, 71). The data that do not support this assumption can potentially be explained by the congruence of maternal and embryonic thermal optimums (72). The second assumption of pregnancy-induced shifts in maternal thermoregulatory precision is often supported (69, 71, 73), but not always (70). An advantage of the MMH over other hypotheses explaining the evolution of squamate viviparity is that it potentially explains the persistence of viviparity in the tropics (69).

The selfish mother hypothesis (SMH) extends the focus of the MMH in squamates from a single reproductive event to lifetime reproductive success (74, 75). It argues that the MMH focuses implicitly on a single reproductive event and contests that under scenarios such as resource limitation or for long-lived species, mothers will favor their survival over clutch success (74, 75). Support for the SMH in the oviparity–viviparity transition is negligible, though it finds reasonable support when viewed more broadly as "selfish maternal effects" (76). The strongest support for the SMH in the transition to viviparity comes from recent climatic niche models of lizards with intraspecific variation in reproductive mode, which support the SMH over both the MMH and CCH (62). Others (e.g., 74, 75, 77) have covered the debate on the MMH and SMH.

Finally, the conflict-driven hypothesis argues that conflict between parents and between mothers and their offspring can explain the transition from oviparity to viviparity across all taxa (78). It argues that males and offspring benefit from increased duration of embryo retention at the cost of mothers and suggests that if embryos can influence the timing of parturition, conflict may select for the evolution of viviparity (78, 79). Prolonged retention of embryos provides an extended window for embryos to exert influence over maternal resource allocation (e.g., thermal resources, nutrient supplementation via matrotrophy). This parent–offspring conflict instigates a coevolutionary dynamic, potentially explaining the diversification of matrotrophic strategies (78). Several recent reviews discuss the conflict hypothesis and its broader applicability to sexual conflict across taxa (36, 78, 79).

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Many of the putative factors resulting in the evolution of internal brooding are similar to viviparity. However, a notable distinction is that brooding likely evolved from increased behavioral investment in parental care (e.g., egg guarding), leading to physiological changes, rather than in viviparity, which is led by physiological processes (8, 80, 81). Environmental challenges such as nest dehydration (amphibians; 81), environmental stress (marine invertebrates; 80), and nest or larval predation (all taxa; 8, 81, 82) may all be factors. In amphibians, breeding in a small rather than large body of water (a proxy for desiccation risk and thermal stability) is often linked to diversification of parental care strategies such as clutch carrying (81), a trait that may be a transitional stage proceeding internal brooding. Fish and amphibians with putative transitional stages of brooding (clutch carrying) could be robust model systems but are underutilized in understanding the evolution of internal brooding. For instance, experimental investigation of clutch carrying in midwife toads (*Alytes obstetricans*) appears to support the role of the MMH in the evolution of this transitional stage, as fathers incur thermal and locomotor costs to improve offspring fitness (83, 84). Such experimental investigations are powerful tools for addressing the adaptive benefits of internal brooding, but they are rare. Both fish and amphibians are effective experimental systems for identifying the costs and benefits of internal brooding given the potential to compare different forms of parental investment across species.

For fish and marine invertebrates, offspring predation during early development is often considered the more likely driver of transitions to brooding (8, 80). Alternatively, brooding in both frogs and marine invertebrates may be a way to increase offspring successfully recruiting into suitable microenvironments (80, 81). In marine invertebrates, other potential advantages of internal brooding include providing developmental stability (similar to MMH predictions) or shielding embryos from pathogens (80). In the case of male internal brooding, increased paternity assurance is a likely adaptive benefit (8, 82, 85). We direct readers to taxon-specific reviews that cover brooding in amphibians (81, 85), invertebrates (80), and syngnathid fish (8) for more information.

Although many questions around the transition from oviparity to viviparity and oviparity to internal brooding have been investigated comprehensively, many outstanding questions remain. Cold climates unquestionably influence the evolution of viviparity and thus have been a focus in viviparity research. However, this focus has led to relatively few investigations of the benefits of viviparity at tropical latitudes (69, 86). Overall, there has been a strong bias toward investigating the CCH and MMH, resulting in many hypotheses being under-investigated (e.g., SMH) or not investigated at all (e.g., nest predation). Adequate investigation of competing hypotheses across taxa is critical, because unsupported ideas can be dismissed more confidently when not applicable to specific taxa (e.g., environmental predictability for squamates). Similar issues are present in research into the evolution of internal brooding, where experimental data are often lacking to support the hypothesis. Finally, most investigations have focused on a univariate approach to the evolution of viviparity and internal brooding, although multiple selective agents and their interaction likely play a role in the evolution of this trait. Better determining how forces interact, their specific targets, and how this shapes the benefit of viviparity or internal brooding in some environments but not others will be an important direction for future research.

PHYSIOLOGY OF VIVIPARITY AND REQUIREMENTS FOR EMBRYONIC DEVELOPMENT

Compared to external embryonic development, many of the physiological challenges of internal incubation of embryos are common across taxa in which viviparity has evolved independently. Several broad categories of convergent adaptations are evident. This section aims to highlight broad patterns of adaptation across viviparous species, focusing on vertebrates rather than providing an exhaustive list of taxa displaying each adaptation.

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Modifications to Parturition Timing

Viviparity and oviparity are fundamentally different in the timing of expulsion of the egg/embryo from the female reproductive tract. In oviparity, some or all embryonic development occurs externally. In viviparity, embryonic development is completed during retention inside the female. Viviparity likely evolved after internal fertilization via gradual increases in the duration of egg retention. Few extant species display intermediate phenotypes along this gradient (see the section titled Transitional Forms of Pregnancy), but they represent critical models for understanding the processes leading to the completion of embryonic development during internal gestation.

Internal Fertilization of Eggs

All viviparous species are necessarily internal fertilizers because embryonic development occurs entirely within the female reproductive tract in this parity mode. This trait is likely one of the most critical exaptations associated with viviparity; once fertilized eggs were associated with the female reproductive tract, increases in the duration of embryo retention would be possible, eventually leading to complete embryonic development in the female reproductive tract (e.g., 18, 22, 23, 36). In support of this hypothesis, viviparity has evolved most often in vertebrate lineages that are ancestrally internal fertilizers, for example, in squamates, poeciliids, and caecilians, which collectively comprise 85% of independent origins of vertebrate viviparity (1, 7, 87). In contrast, viviparity has evolved only rarely in teleosts and anurans; internal fertilization is a derived feature in these lineages (1, 7). However, the correlation between internal fertilization and viviparity is not perfect: All birds (no origins of viviparity) and most urodeles (two origins of viviparity) are internal fertilizers (1, 87).

In vertebrates, the mechanisms facilitating internal fertilization vary. For example, intromittent organs are present in viviparous (and many oviparous) amniotes (penis or hemipenis) and in chondrichthyans (modified pelvic fins knowns as claspers). Caecilians have a phallodeum (87), whereas viviparous teleost fishes have a diverse array of male intromittent organs, including gonopodia (modified anal fins) and modified genital papillae (e.g., 88, 89). Many viviparous species achieve internal fertilization of eggs without a male intromittent organ: Cloacal transfer of semen into the female reproductive tract occurs in viviparous anurans (87) and likely in coelacanths (90), whereas spermatophore transfer is present in viviparous urodeles (87).

Exchange of Respiratory Gases

Embryonic development requires a steady supply of oxygen and removal of carbon dioxide, the demand for which accelerates as development progresses. In oviparous embryos, respiratory gases are exchanged with the surrounding water, air, or air pockets in the surrounding soil through a permeable egg covering (e.g., 91), which may be enhanced by parental care behaviors such as egg fanning (e.g., 92). However, once embryos begin to undergo substantial development inside the mother's body, respiratory gas exchange becomes a major physiological challenge because O_2/CO_2 diffuse very slowly through animal tissues (93). Therefore, it is predictable that a plethora of adaptations have emerged that circumvent this problem. In many cases, these adaptations also fulfil other exchange functions; nutrients (see the section titled Transport of Nutrients from Parent to Embryo) are frequently transported to developing embryos via adaptations that may have originally functioned in respiratory gas exchange (1).

Respiratory gas exchange is impeded by physical barriers between the embryo and the external environment, including any egg coverings. The thicker the egg covering, the slower the diffusion (following Fick's first law of diffusion). Therefore, viviparity and indeed extended internal egg

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retention (transitional forms of pregnancy) are associated with the complete loss of or dramatic reduction in egg covering thickness or complexity (e.g., 9, 20, 23, 94–96).

Another common adaptation for respiratory gas exchange that has evolved convergently in viviparous species is the dramatic remodeling of maternal gestational tissues during pregnancy. Depending on the species, this remodeling can take the form of an expanded maternal gestational surface area, for example, by the development of villi, folds, and lamellae (e.g., 23, 97), or a reduction in the diffusion distance between the embryonic and maternal circulation, which increases the rate of exchange (reviewed in 9). Expansion of the gestational tissue vascular bed during pregnancy has also evolved frequently in various viviparous vertebrates (e.g., 9, 98, 99). Placentae also develop via extensive remodeling of gestational tissue architecture and vascularity (see the section titled Placentation).

Elegant evolutionary solutions to the physiological challenge of respiratory gas exchange are also exemplified in embryonic adaptations. Modifications to embryonic biochemistry can produce increased embryonic blood oxygen affinity, increasing the oxygen transfer rate from maternal to embryonic blood (e.g., 100, 101). Changes to embryonic morphology can also facilitate oxygen uptake; for example, tissue hypertrophy produces embryonic gills, fins, trophotaeniae (gut extensions), and other structures with large surface area for respiratory gas exchange with the gestational environment (e.g., 102, 103).

Finally, behavioral adaptations of both viviparous mothers and embryos can increase the supply of respiratory gases for development. These mechanisms appear to have evolved primarily in chondrichthyans. Pregnant mothers of some species flush fresh seawater into the uterus (e.g., 104). Some embryos use rhythmic buccal pumping to flush fluid across their respiratory tissues (e.g., 105), and in one shark, embryos swim between paired uteri and can protrude their heads through the cervix, presumably exposing them to fresh seawater (106).

Immunoregulation

Embryos incubated internally, particularly without egg coverings, are closely associated with parental tissues. In viviparity, this situation results in mothers being exposed to embryonic antigens inherited from the father. This potentially creates an immunological conundrum (e.g., 107, 108), particularly in placental species, where parental and embryonic tissues are intimately associated. Mammalian pregnancy is best understood in this context; we know that the attachment reaction in marsupial pregnancy produces cooperative inflammation and subsequently birth, whereas in eutherian mammals, harmful inflammation is reduced, allowing embryo implantation, extended gestation, and finally inflammation at parturition (reviewed in 109, 110). Interested readers are directed to recent reviews of mammalian immunity in pregnancy (e.g., 111).

In contrast to mammalian pregnancy, our knowledge of immunoregulation in other viviparous taxa is limited. The immune environment in gestational tissues seems to be regulated during squamate pregnancy (e.g., 112–114), and immune function may be reduced (e.g., 115). Immune gene expression is regulated (116), and immune components (117) and inflammatory mediators (e.g., 118) are present in the gestational tissues of several sharks during pregnancy. Limited research in amphibians suggests that immune regulation may also be necessary during pregnancy in this lineage (119, 120).

Transport of Nutrients from Parent to Embryo

The dominant vertebrate strategy for providing embryos with nutrients for growth and development is lecithotrophy (provision of all embryonic nutrition in the egg yolk before ovulation). However, the close association between mother and embryos in viviparous taxa provides the

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Table 1 Taxonomic distribution of different modes of parentotrophy in extant viviparous vertebrates and extant oviparous, brooding vertebrates[a]

Lineage	Placentotrophy	Histotrophy/ histophagy	Oophagy	Embryophagy	Matrophagy
Chondrichthyes	X	X	X	X	
Actinopterygii	X	X	X	X	
Coelacanthiformes	Possible		Possible		
Lissamphibia	X	X	X	X	X
Squamata	X				
Mammalia	X				

[a]Modified from Blackburn (1) with information from References 1, 25, 36, and references therein. Further research is required to definitively determine the mode(s) of nutrient transport in coelacanths (39). X indicates presence of each mode of extraembryonic nutrition in each lineage.

opportunity for substantial post-ovulation transport of nutrients from mothers to embryos (matrotrophy) in some species. Importantly, lecithotrophy and matrotrophy are not mutually exclusive: Some taxa utilize both mechanisms of embryonic nutrition to varying degrees. Matrotrophy has evolved independently in viviparous vertebrates 33 or more times, with most origins in fishes and amphibians (1).

The mechanisms enabling matrotrophy are diverse, especially in teleost fishes and amphibians (**Table 1**). Viviparous animals may utilize histotrophy/histophagy, oophagy, embryophagy (embryonic cannibalism), matrophagy, placentotrophy, or a combination of these modes. For example, the gray nurse shark (*Carcharias taurus*) famously exhibits embryophagy but also employs histotrophy and oophagy at some stages of pregnancy (121).

Placentotrophy. Placentotrophy (transport of nutrients across the placental interface) is observed in all viviparous mammals. In viviparous squamates, water, respiratory gas, and calcium transport commonly occur across the placenta of lecithotrophic species (e.g., 122). Additionally, substantial placentotrophy, in which significant quantities of other nutrients are provided placentally, has evolved in six clades of scincid lizard (1). Substantial placentotrophy has also evolved in viviparous teleosts; for example, Pacific four-eyed fish (*Anableps dowii*) embryos increase in dry mass more than 8,000 times via placental nutrient transport (123). In chondrichthyans, substantial placentotrophy is present in a single clade of sharks, Carcharhiniformes (36). Placentotrophy is not known in viviparous amphibians, although there is a likely origin of placentotrophy in oviparous, brooding frogs (see the section titled Convergence of Viviparity and Oviparity with Brooding across Vertebrates).

Histotrophy and histophagy. Histotrophe refers to nutritive secretions from gestational tissues, which may be either absorbed (histotrophy) or ingested (histophagy) by developing embryos; these nutritive mechanisms probably occur simultaneously (23). Histotrophy is also likely to co-occur with other modes of embryonic nutrition. Histotrophe provides embryonic nutrition post-ovulation in some amphibians, chondrichthyans, and actinopterygians (e.g., 1, 36, 87). In some species, mothers/embryos exhibit morphological specializations to increase the secretory/absorptive surface area for histotrophy, for example, uterine projections (trophonemata) or embryonic extensions (e.g., trophotaeniae) (23).

Oophagy. Oophagy refers to ingesting fertilized sibling eggs, or unfertilized nurse eggs, which may be eaten whole or fragmented. Oophagy is present in several groups of amphibians and chondrichthyans and in three teleost lineages (e.g., 9, 23, 121). For embryos to engage in oophagy, they must have functional digestive systems; teeth, jaw apparatus, and digestive tract modifications may develop precocially in these species (1).

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Embryophagy. Ingestion of siblings during gestation is termed embryophagy, adelphophagy, or embryonic cannibalism and is often associated with oophagy (1). This mechanism of matrotrophy is quite rare. It has been described in only one shark [*C. taurus* (121)], two teleost fishes [*Cataetyx laticeps* and *Nomorhamphus ebrardtii* (124, 125)], and two amphibians [*Salamandra salamandra* and *Typhlonectes compressicauda* (126, 127)].

Matrophagy. Matrophagy is another relatively rare strategy enabling embryonic nutrition to be provided post-ovulation. It has evolved four times independently in caecilians (Gymnophiona) and once in the salamander *Salamandra atra* (e.g., 1, 9, 87). Precocial development of embryonic digestive systems, including tiny deciduous teeth, allows matrophagic embryos to scrape off and ingest specialized uterine epithelial tissue and secretions (9, 87).

Challenges in matrotrophy research. There are likely many examples of matrotrophy, particularly histotrophy, that may be challenging to detect but have yet to be discovered. Matrotrophy is often identified by comparing dry mass of newborn embryos to unfertilized or newly fertilized eggs, producing a "matrotrophy index" (e.g., 122, 128). The rationale is that if newborn embryos are heavier than newly fertilized eggs, then additional nutrients must have been provided to embryos during gestation. Similarly, if no dry mass is lost, matrotrophy is also indicated because growth and metabolism consume dry mass in strictly lecithotrophic embryos. However, a challenge in matrotrophy research is identifying species that display incipient (limited) matrotrophy; these species would be expected to lose less mass than a lecithotrophic embryo. The amount of dry mass lost during lecithotrophic embryonic development is debated (possibly ranging from ~20% to 40%), but it probably differs depending on the taxonomic group. Ideally, one would compare dry mass loss with a closely related species that was strictly lecithotrophic [i.e., oviparous; oviparity with matrotrophy is rare, although present in monotremes (5)]. Because there is likely no universal baseline matrotrophy index that implies certain matrotrophy, additional methods should be used to determine matrotrophy in suspected incipient matrotrophs. Compositional analysis of macronutrients, tracing transport of radio- or stable-isotope-labeled nutrients such as amino acids, and morphological analysis to identify secretory structures in gestational tissue would all be suitable alternatives (e.g., 1, 129, 130).

CONVERGENCE OF VIVIPARITY AND OVIPARITY WITH BROODING ACROSS VERTEBRATES

In vertebrates, oviparity with brooding is exhibited only by teleost fishes and amphibians. In these species, eggs are incubated inside or on the surface of the body of one of the parents at locations other than the female reproductive tract, in a situation analogous to viviparity. Brooding in enclosed gestational tissues includes skin pouches [e.g., in marsupial frogs (*Gastrotheca* spp.) and seahorses (*Hippocampus* spp.)] and buccal cavities [although these may open periodically during gestation, e.g., in some mouthbrooding cichlid fishes (92)]. Semi-enclosed gestational tissues in oviparous brooding species include pockets, fins, and skin outgrowths. For example, female ghost pipefish (Solenostomidae) brood their developing embryos on stalklike outgrowths of the integument (cotylephores), inside fused pelvic fins (131), and male nurseryfish (*Kurtus gulliveri*) brood embryos on a vascularized forehead hook (132).

Internal or semi-internal incubation of embryos at sites other than the female reproductive tract creates a similar set of physiological challenges during embryonic development as in viviparity. In particular, oviparous brooding species with fully enclosed gestational tissues are likely to have adaptations facilitating parent–embryo respiratory gas exchange, including expansions of the vascular bed in gestational tissues, reduced or absent egg coverings that would otherwise be a

barrier to molecular exchange, and/or expanded surface area of gestational or embryonic tissues; some of these adaptations may be coopted for nutrient transport (1, 25, 133).

Although many oviparous brooding species are lecithotrophic, there is the opportunity for nutrient supplementation from the brooding parent, either the mother (matrotrophy) or father (patrotrophy), collectively termed parentotrophy (**Table 1**). As discussed in the section titled Placentation, some oviparous brooders have placentae. Whereas the parental portion of the placenta in viviparous species is always maternal ovarian tissue or maternal uterus/oviduct, this constraint is removed in oviparous brooding species, and placentae result from the close association of other tissues in either parent (25). For example, male seahorses (*Hippocampus* spp.) have placentae composed of paternal brood pouch epithelium, which transport respiratory gases, lipids, and proteins to developing embryos (134–136). Within skin-brooding anurans, female *Gastrotheca riobambae* and *Gastrotheca excubitor* incubate embryos in dorsal integumentary pouches; embryos have vascularized saclike gill extensions that appose to the mother's pouch tissues, facilitating respiratory gas exchanges and likely nutrient transport (137, 138). Histotrophy/histophagy are likely in male *Rhinoderma darwinii* frogs, which brood embryos in the vocal sac (139). Oviparous brooding has not been the focus of as much research as viviparity, but it is a promising new area in understanding the proximate and ultimate mechanisms underlying the evolution of internal incubation.

VIVIPARITY, BROODING, AND MOLECULAR CONVERGENCE

A major focus of current viviparity research is understanding whether viviparity (and indeed oviparity with brooding) and associated convergent adaptations (e.g., placentotrophy) evolve predictably. Some studies have compared independent origins of viviparity within closely related lineages. For example, genomic analysis in a subset of viviparous poeciliid fishes with six independent origins of placentotrophy identified convergent shifts in the evolutionary rate of coding regions of the genome and differential conservation of noncoding elements associated with the repeated evolution of placentotrophy (140).

Other studies have identified similar genes expressed in gestational tissues during pregnancy in a range of taxa, including poeciliid fishes and mammals (e.g., 141) and viviparous lizards and mammals (e.g., 112). Similarities in gene expression in gestational tissues have also been detected between oviparous brooding and viviparous species, including male-pregnant syngnathid fishes and several origins of viviparity in amniotes (e.g., 15, 142, 143). Although these similarities in pregnancy across distantly related vertebrates are intriguing, many studies have not quantified whether the genetic overlap between independent origins of the focal trait is more than would be expected by chance. Therefore, their ability to determine whether the evolution of viviparity is indeed predictable at the genetic level is limited.

More recent, quantitative analyses of overlapping gene use have failed to find evidence of convergent genetic mechanisms underpinning independent origins of viviparity. For example, analyses of uterine gene expression in viviparous mammals, lizards, and a shark detected no significant overlap in gene expression in either viviparity or placentotrophy (116). Similarly, a genomic study of two origins of viviparity in poeciliid and goodeid fishes detected no significant molecular convergence in protein-coding genes (144). Analyses of amino acid substitution in genes involved across multiple origins of squamate viviparity found a level of convergent amino acid replacement that could be explained by chance (114). Although a large-scale genomic study of viviparity across vertebrates identified contraction/expansion of 15 protein families that were associated with viviparity in some species and noncoding and protein-coding regions with accelerated substitution rates in a subset of viviparous lineages, no convergent changes were

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found across all viviparous taxa (145). A mounting body of evidence, therefore, suggests that despite some genetic signatures associated with a subset of independent origins of viviparity, multiple distinct evolutionary and genetic pathways likely have produced viviparity and brooding in vertebrates. The genes recruited to expression in gestational tissues when pregnancy evolves are likely to be influenced by the cellular origins of the gestational tissues and evolutionary history of the lineage, which may dictate the availability of genes upon which selection can act.

MOLECULAR TOOLS TO UNDERSTAND THE BIOLOGY AND EVOLUTION OF PREGNANCY

We predict that further insights into the molecular evolution of pregnancy will result from incorporating new technologies into studies of vertebrate pregnancy. RNAseq is used to assess gene expression in gestational tissues and infer biological function and is already widespread (e.g., 110, 112, 141, 142). Taken further, ancestral transcriptome reconstruction allows us to trace the evolution of gestational gene expression across a lineage (e.g., 146). The reduced cost of genome sequencing technologies and availability of methods to study gestational tissue epigenomes [e.g., ATAC seq (147)] also mean that it is now possible to explore gene regulation across independent origins of pregnancy in non-model species. Single-cell sequencing has helped to identify individual cell type functions within mammalian placentae (reviewed in 148) but has not applied more broadly to vertebrate pregnancy. This method will likely be particularly useful in understanding the biology of early pregnancy, including embryo–parent communication and implantation of the conceptus.

Such techniques will likely be most useful in non-model species after broad-scale investigations of gestational tissue and placental biology using other techniques, including RNAseq, to understand gestational gene expression at a larger scale. Their application to species with transitional forms of pregnancy is likely to lead to significant advances in our understanding of how pregnancy evolves at a molecular level by providing rare information about intermediate states in the evolution of pregnancy (4). We also highlight that well-established physiological and histological techniques should not be neglected, particularly in species in which the biology of pregnancy is poorly understood (25). For example, techniques including advanced electron microscopy and three-dimensional imaging, such as microcomputed tomography, can be used to identify gestational and placental structures and infer function (reviewed in 25, 149). In tandem with physiological techniques such as compositional analysis, these tools can identify the source of embryonic nutrition during pregnancy or identify the hormonal control of gestation and parturition (e.g., 150). This fundamental biological knowledge and information on the molecular basis of gestation can then be incorporated into models working to understand how and why pregnancy has evolved.

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